

Project 1

Masum Billah, Madhu Chencharapu, Gabriel Loos, Brendan McDonnell, Roshan Ravichandran

February 2026

1 Introduction

In this project, we will explore three different datasets using Neural Networks (2, 3, and 4 layers). The three datasets we will be using are:

- **Auto MPG:** This dataset contains information about different cars. The goal is to predict the miles per gallon (MPG) given the other characteristics of the car. The dataset can be found at <https://archive.ics.uci.edu/dataset>.
- **California Housing Prices:** This dataset contains information about different housing locations. The goal is to predict the median house price of the area given the other characteristics. The dataset can be found at <https://www.kaggle.com/datasets/camnugent/california-housing-prices>.
- **Insurance Charges:** This dataset contains information about different primary beneficiaries of health insurance. The goal is to predict the primary beneficiaries insurance charges given their other characteristics. The dataset can be found at <https://www.kaggle.com/datasets/mirichoi0218/insurance>.

2 Auto MPG

2.1 2 Layer Neural Network

In Scallation, the best results were found when using the reLU as the activation function, a learning rate of 0.001, and a batch size of 1. Since the Auto MPG dataset is relatively small, picking a batch size of 1 does not take much time, so is worth the better results, which are shown in Table 1. In PyTorch, the best results were found when using the reLU as the activation function, a learning rate of 0.01, and a batch size of 32. The results are shown in Table 2. Plots for In-Sample and Out-of-Sample testing can be found in Figure 1.

Table 1: Scallation - Auto MPG 2L NN

Metric	In-Sample	80-20 Split
rSq	0.727231	0.836932
rSqBar	0.722259	0.833959
sst	23819.0	4731.23
sse	6497.08	771.514
sde	3.39572	3.16498
mse0	16.5742	9.89121
rmse	4.07114	3.14503
mae	3.31165	2.36625
smape	14.4265	10.7664
m	392.000	78.0000
dfr	7.00000	7.00000
df	384.000	384.000
fStat	146.255	281.549
aic	-1105.39	-184.051
bic	-1073.62	-165.198

Table 2: PyTorch - Auto MPG 2L NN

Metric	In-Sample	80-20 Split
rSq	0.8232	0.8391
rSqBar	0.8199	0.8232
sst	23818.9935	4910.2699
sse	4211.9875	790.2830
sde	3.3119	3.3363
mse0	10.7449	10.0036
rmse	3.2779	3.1628
mae	2.4941	2.4081
smape	11.4175	11.1501
m	392.0000	79.0000
dfr	8.0000	8.0000
df	384.0000	71.0000
fStat	223.4423	46.2681
aic	946.7758	197.9325
bic	978.5459	216.8881

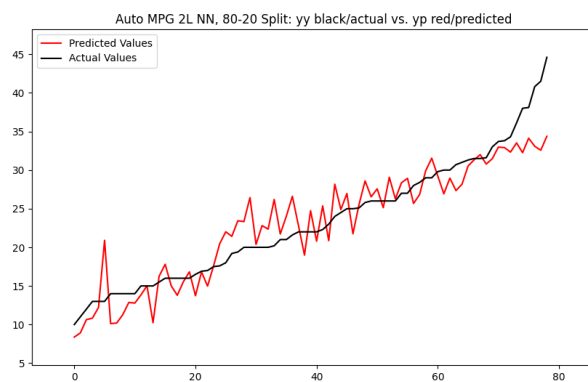
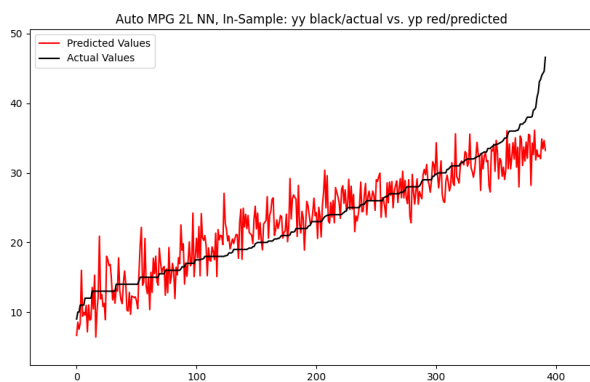
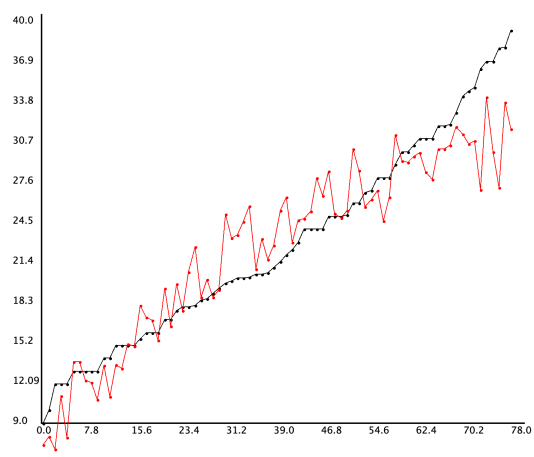
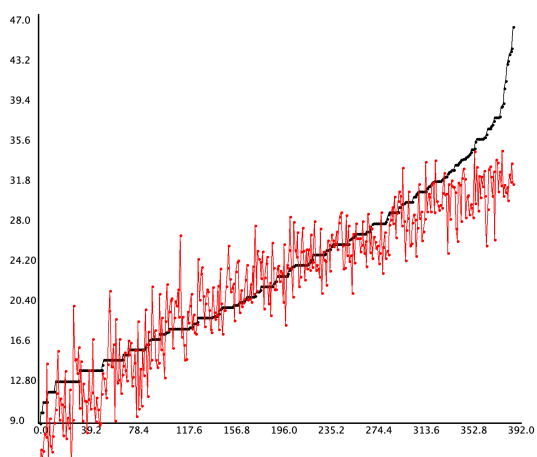


Figure 1: Auto MPG 2 Layer Neural Network Plots
Top Left: Scalation In-Sample, Top Right: Scalation Out-of-Sample
Bottom Left: PyTorch In-Sample, Bottom Right: PyTorch Out-of-Sample
black: actual y-values, red: predicted y-values

2.2 3 Layer Neural Network

In Scallation, the best results were found when using sigmoid as the activation function for the hidden layer, reLU as the activation function for the output layer, $3 \times x.dim2 = 24$ nodes in the hidden layer, a learning rate of 0.01, and a batch size of 8. The results are shown in Table 3. In PyTorch, the best results were found when using LeakyReLU as the activation function for the hidden layer, identity as the activation function for the output layer, 100 nodes in the hidden layer, a learning rate of 0.01, and a batch size of 32. The results are shown in Table 4. Plots for In-Sample and Out-of-Sample testing can be found in Figure 2.

Table 3: Scallation - Auto MPG 3L NN

Metric	In-Sample	80-20 Split
rSq	0.930275	0.868516
rSqBar	0.928822	0.865777
sst	23819.0	4731.23
sse	1660.78	622.081
sde	2.05245	2.79802
mse0	4.23669	7.97540
rmse	2.05832	2.82407
mae	1.50903	2.03520
smape	6.54887	8.62759
m	392.000	78.0000
dfr	8.00000	8.00000
df	384.000	384.000
fStat	640.418	317.064
aic	-821.212	-173.675
bic	-785.471	-152.464

Table 4: PyTorch - Auto MPG 3L NN

Metric	In-Sample	80-20 Split
rSq	0.8855	0.9095
rSqBar	0.8835	0.9005
sst	23818.9935	4910.2699
sse	2726.1212	444.5322
sde	2.6644	2.5022
mse0	6.9544	5.6270
rmse	2.6371	2.3721
mae	1.8997	1.6043
smape	8.0857	7.0419
m	392.0000	79.0000
dfr	8.0000	8.0000
df	384.0000	71.0000
fStat	371.3914	89.1576
aic	776.2343	152.4784
bic	808.0044	171.4340

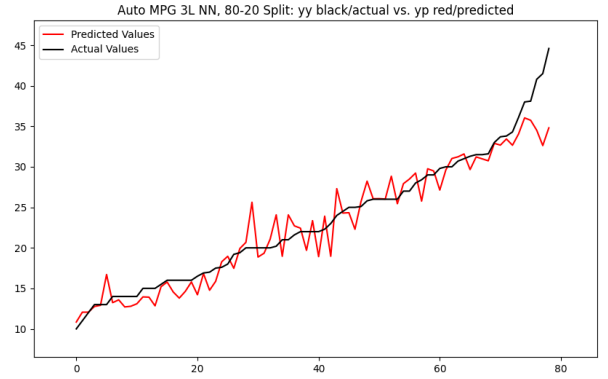
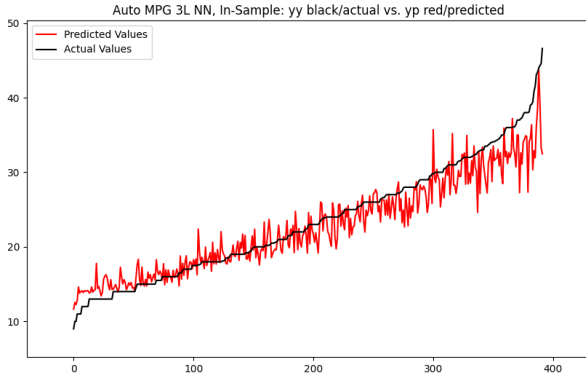
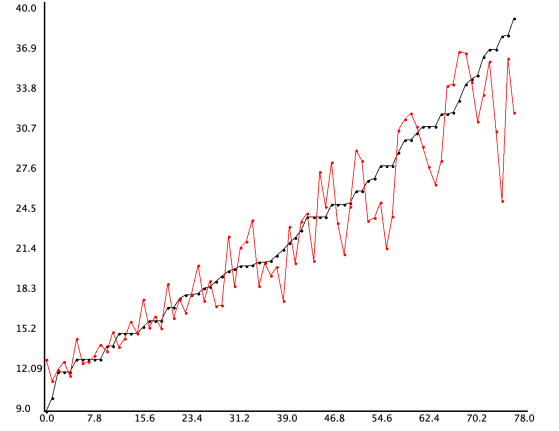
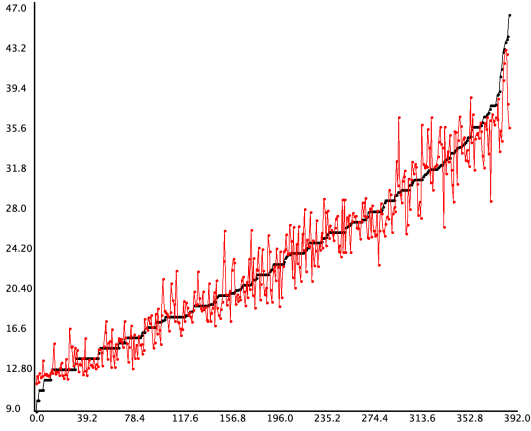


Figure 2: Auto MPG 3 Layer Neural Network Plots
 Top Left: Scalation In-Sample, Top Right: Scalation Out-of-Sample
 Bottom Left: PyTorch In-Sample, Bottom Right: PyTorch Out-of-Sample
 black: actual y-values, red: predicted y-values

2.3 4 Layer Neural Network

3 California House Prices

3.1 2 Layer Neural Network

In Scalation, the best results were found when using the reLU as the activation function, a learning rate of 25, and a batch size of 32. We note the unusually high learning rate that was required in order to get good results. This is most likely due to the y-values having a large scale. The results are shown in Table 5. In PyTorch, the best results were found when using the reLU as the activation function, a learning rate of 0.01, and a batch size of 32. The results are shown in Table 6. Plots for In-Sample and Out-of-Sample testing can be found in Figure 3.

Table 5: Scalation - California House Prices 2L NN

Metric	In-Sample	80-20 Split
rSq	0.633776	0.630655
rSqBar	0.633579	0.630456
sst	2.72264e+14	5.31945e+13
sse	9.97097e+13	1.96471e+13
sde	69653.2	69346.2
mse0	4.87984e+09	4.80840e+09
rmse	69855.8	69342.6
mae	51821.3	50737.2
smape	26.8854	26.5678
m	20433.0	4086.00
dfr	11.0000	11.0000
df	20421.0	20421.0
fStat	3212.73	3169.89
aic	-256883	-51319.7
bic	-256788	-51243.9

Table 6: PyTorch - California House Prices 2L NN

Metric	In-Sample	80-20 Split
rSq	0.6468	0.6516
rSqBar	0.6466	0.6506
sst	272264434648930.1875	54723646809444.8438
sse	96172625731084.4375	19066274074331.2266
sde	68625.7706	68402.0487
mse0	4706730569.7198	4665102538.3732
rmse	68605.6162	68301.5559
mae	49654.9430	50131.4253
smape	26.7647	26.6343
m	20433.0000	4087.0000
dfr	12.0000	12.0000
df	20421.0000	4075.0000
fStat	3115.8995	635.0821
aic	455113.0754	91014.4163
bic	455208.1743	91090.2031

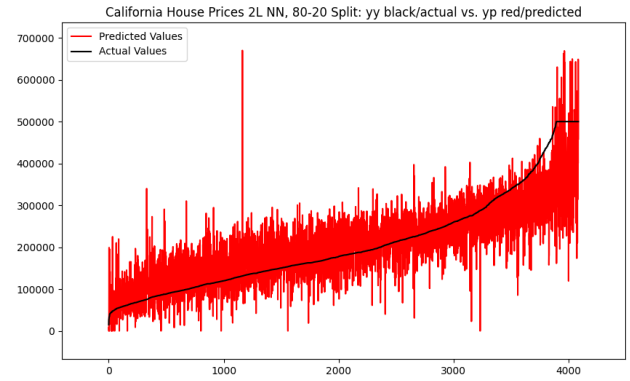
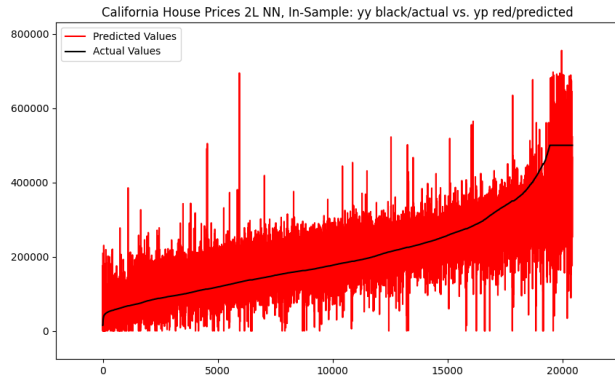
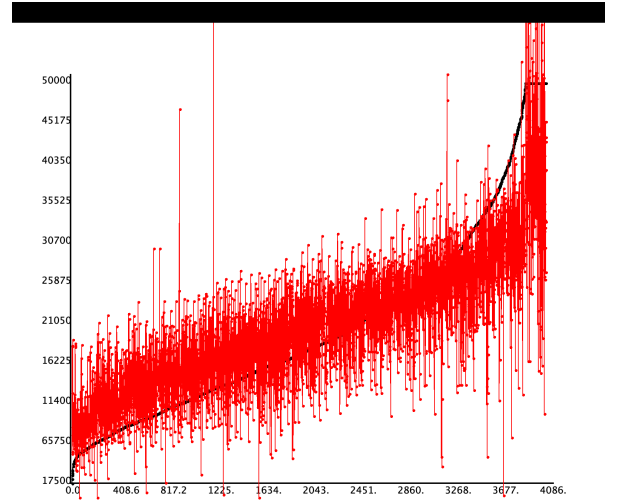
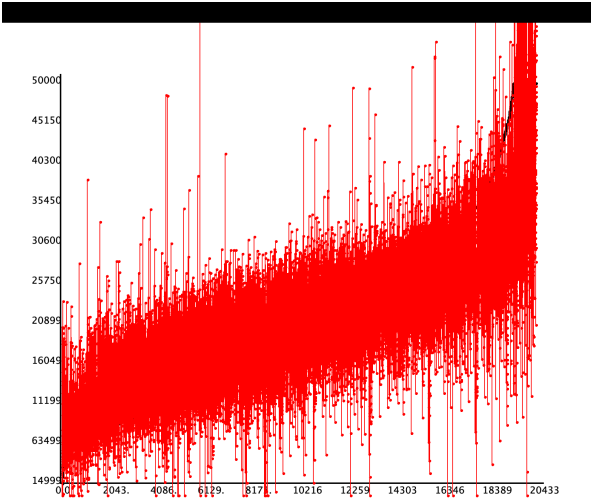


Figure 3: California House Prices 2 Layer Neural Network Plots
Top Left: Scalation In-Sample, Top Right: Scalation Out-of-Sample
Bottom Left: PyTorch In-Sample, Bottom Right: PyTorch Out-of-Sample
black: actual y-values, red: predicted y-values

3.2 3 Layer Neural Network

In Scalation, the best results were found when using `lreLU` as the activation function for the hidden layer, identity as the activation function for the output layer, $3 \times x.dim2 = 36$ nodes in the hidden layer, a learning rate of 0.01, and a batch size of 32. The results are shown in Table 7. In PyTorch, the best results were found when using `LeakyReLU` as the activation function for the hidden layer, identity as the activation function for the output layer, 100 nodes in the hidden layer, a learning rate of 0.01, and a batch size of 32. The results are shown in Table 8. Plots for In-Sample and Out-of-Sample testing can be found in Figure 4.

Table 7: Scalation - California House Prices 3L NN

Metric	In-Sample	80-20 Split
rSq	0.779769	0.756155
rSqBar	0.779639	0.756012
sst	27226.4	5319.45
sse	5996.11	1297.12
sde	0.541323	0.562612
mse0	0.293452	0.317454
rmse	0.541712	0.563431
mae	0.371532	0.389204
smape	19.3019	20.5293
m	20433.0	4086.00
dfr	12.0000	12.0000
df	20421.0	20421.0
fStat	6025.36	5277.08
aic	-16441.3	-3427.61
bic	-16338.3	-3345.51

Table 8: PyTorch - California House Prices 3L NN

Metric	In-Sample	80-20 Split
rSq	0.8057	0.7961
rSqBar	0.8056	0.7956
sst	272264434648930.1250	54723646809444.8281
sse	52901444882088.6094	11156430483920.8633
sde	50897.3609	52323.7456
mse0	2589019961.9287	2729735865.8970
rmse	50882.4131	52246.8742
mae	34838.6035	35574.6590
smape	18.1115	18.1492
m	20433.0000	4087.0000
dfr	12.0000	12.0000
df	20421.0000	4075.0000
fStat	7056.5363	1326.1142
aic	442899.9831	88824.1727
bic	442995.0819	88899.9595

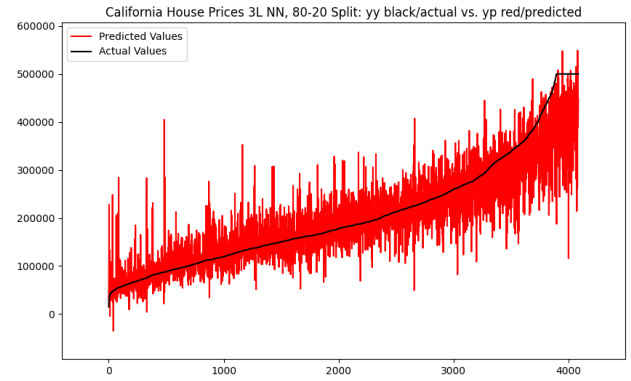
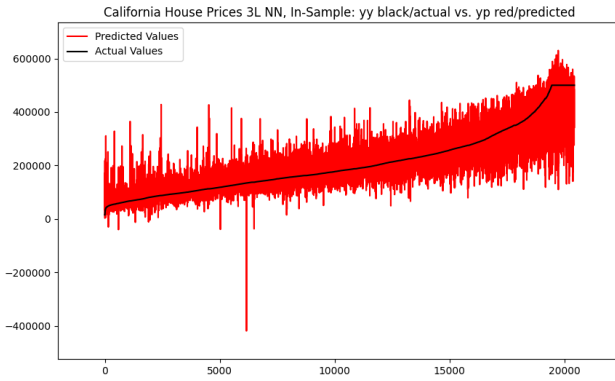
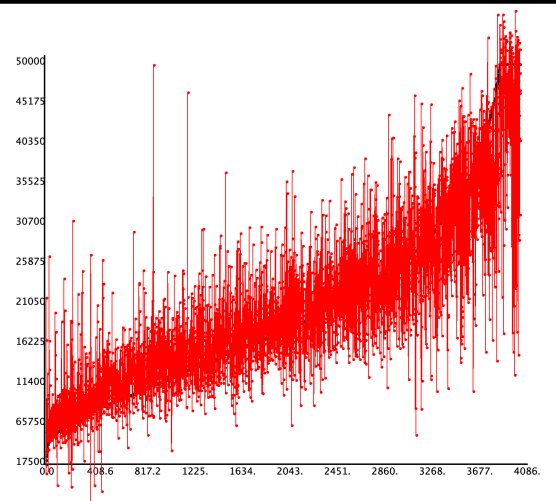
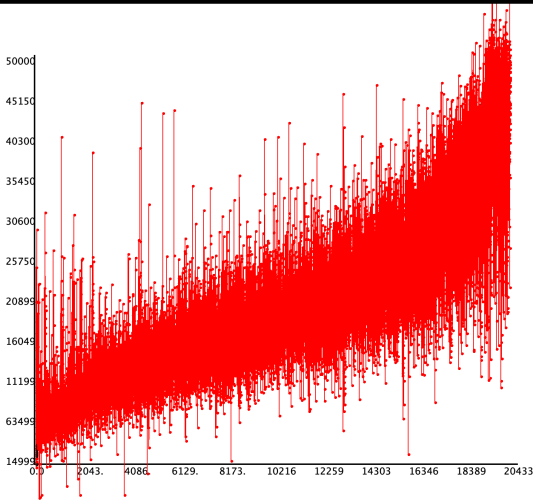


Figure 4: California House Prices 3 Layer Neural Network Plots
Top Left: Scalation In-Sample, Top Right: Scalation Out-of-Sample
Bottom Left: PyTorch In-Sample, Bottom Right: PyTorch Out-of-Sample
black: actual y-values, red: predicted y-values

3.3 4 Layer Neural Network

4 Insurance Charges

4.1 2 Layer Neural Network

In Scalation, the best results were found when using the reLU as the activation function, a learning rate of 200, and a batch size of 32. We note the unusually high learning rate that was required in order to get good results. This is most likely due to the y-values having a large scale. The results are shown in Table 9. In PyTorch, the best results were found when using the reLU as the activation function, a learning rate of 0.01, and a batch size of 32. The results are shown in Table 10. Plots for In-Sample and Out-of-Sample testing can be found in Figure 5.

Table 9: Scalation - Insurance Charges 2L NN

Metric	In-Sample	80-20 Split
rSq	0.722979	0.682065
rSqBar	0.721521	0.680392
sst	1.96074e+11	4.06432e+10
sse	5.43167e+10	1.29219e+10
sde	6346.14	6966.63
mse0	4.05954e+07	4.83965e+07
rmse	6371.45	6956.76
mae	4519.55	4729.99
smape	39.5992	41.9288
m	1338.00	267.000
dfr	7.00000	7.00000
df	1330.00	1330.00
fStat	495.869	407.607
aic	-13602.9	-2725.13
bic	-13561.3	-2696.43

Table 10: PyTorch - Insurance Charges 2L NN

Metric	In-Sample	80-20 Split
rSq	0.7516	0.8007
rSqBar	0.7503	0.7953
sst	196074221568.3671	42646829902.3297
sse	48702200251.0962	8498989265.5920
sde	6051.2970	5717.3788
mse0	36399252.8035	31712646.5134
rmse	6033.1793	5631.3983
mae	4158.4738	3932.9280
smape	39.1264	36.6337
m	1338.0000	268.0000
dfr	8.0000	8.0000
df	1330.0000	260.0000
fStat	503.0696	130.5808
aic	23310.6587	4644.9566
bic	23352.2501	4673.6845

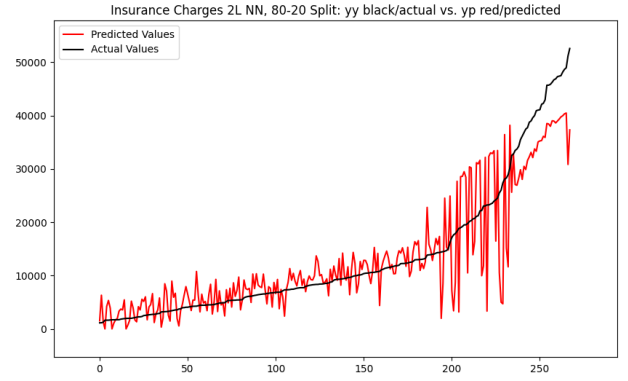
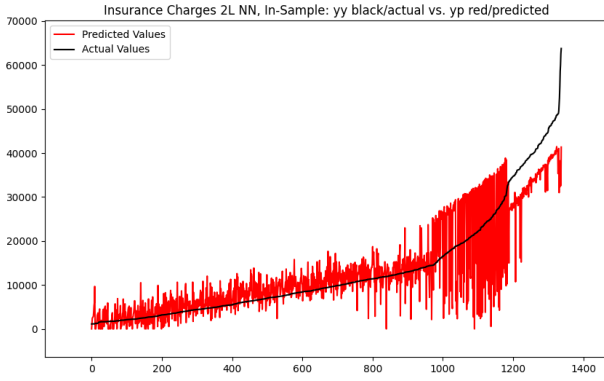
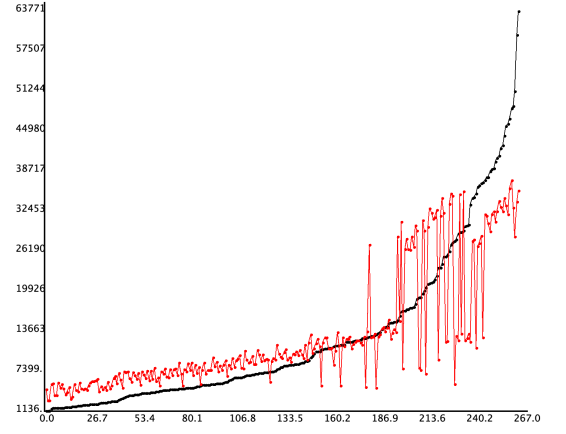
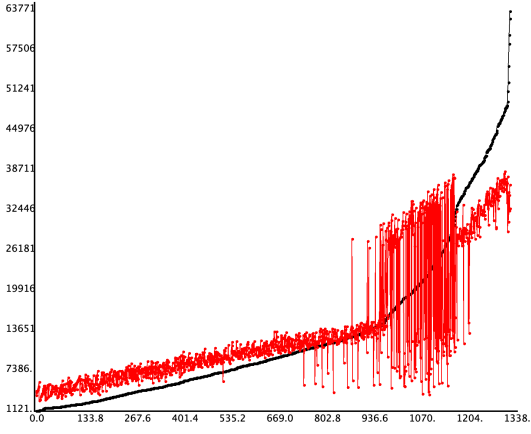


Figure 5: Insurance Charges 2 Layer Neural Network Plots
Top Left: Scalation In-Sample, Top Right: Scalation Out-of-Sample
Bottom Left: PyTorch In-Sample, Bottom Right: PyTorch Out-of-Sample
black: actual y-values, red: predicted y-values

4.2 3 Layer Neural Network

In Scalation, the best results were found when using `lrelu` as the activation function for the hidden layer, identity as the activation function for the output layer, $2 \times x.dim2 = 16$ nodes in the hidden layer, a learning rate of 0.01, and a batch size of 16. The results are shown in Table 11. In PyTorch, the best results were found when using `LeakyReLU` as the activation function for the hidden layer, identity as the activation function for the output layer, 100 nodes in the hidden layer, a learning rate of 0.01, and a batch size of 32. The results are shown in Table 12. Plots for In-Sample and Out-of-Sample testing can be found in Figure 6.

Table 11: Scalation - Insurance Charges 3L NN

Metric	In-Sample	80-20 Split
rSq	0.867231	0.817677
rSqBar	0.866432	0.816580
sst	5446.51	1128.98
sse	723.127	205.838
sde	0.735248	0.877561
mse0	0.540454	0.770930
rmse	0.735156	0.878026
mae	0.416711	0.534961
smape	24.6632	31.2906
m	1338.00	267.000
dfr	8.00000	8.00000
df	1330.00	1330.00
fStat	1085.92	745.594
aic	-1468.87	-326.127
bic	-1422.08	-293.842

Table 12: PyTorch - Insurance Charges 3L NN

Metric	In-Sample	80-20 Split
rSq	0.8766	0.8891
rSqBar	0.8760	0.8861
sst	196074221568.3671	42646829902.3297
sse	24191406710.2515	4730156280.8836
sde	4264.8596	4265.3146
mse0	18080274.0734	17649836.8690
rmse	4252.0906	4201.1709
mae	2466.8147	2735.5215
smape	23.6520	27.4890
m	1338.0000	268.0000
dfr	8.0000	8.0000
df	1330.0000	260.0000
fStat	1181.2260	260.5182
aic	22374.4243	4487.9115
bic	22416.0158	4516.6394

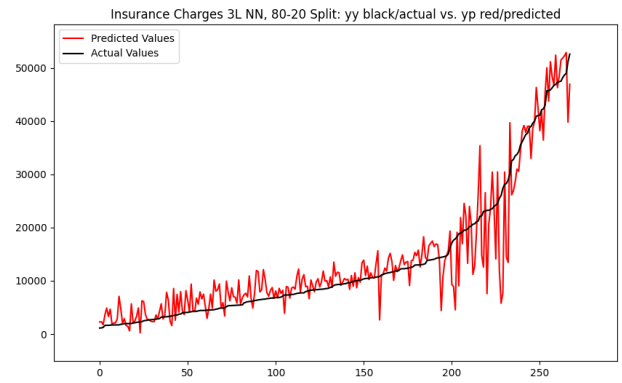
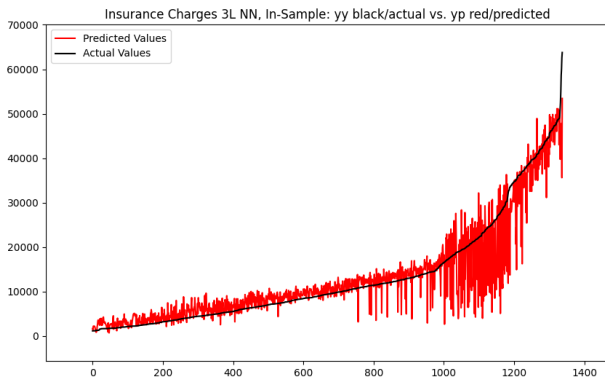
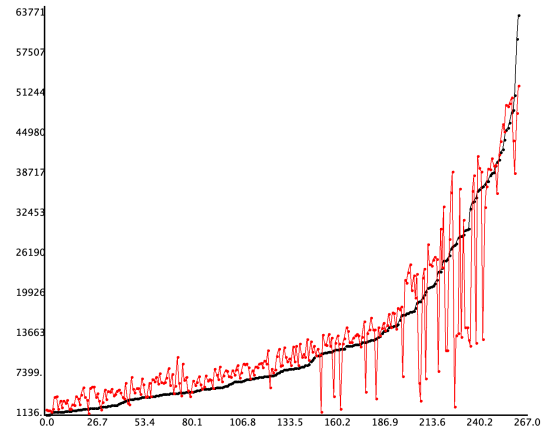
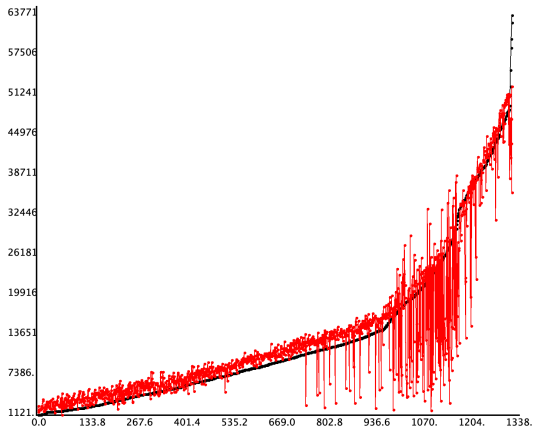


Figure 6: Insurance Charges 3 Layer Neural Network Plots
 Top Left: Scalation In-Sample, Top Right: Scalation Out-of-Sample
 Bottom Left: PyTorch In-Sample, Bottom Right: PyTorch Out-of-Sample
 black: actual y-values, red: predicted y-values

4.3 4 Layer Neural Network

5 Summary

5.1 Auto MPG

Table 13: Scalation: - Auto MPG In-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.727231	0.930275
rSqBar	0.722259	0.928822
sst	23819.0	23819.0
sse	6497.08	1660.78
sde	3.39572	2.05245
mse0	16.5742	4.23669
rmse	4.07114	2.05832
mae	3.31165	1.50903
smape	14.4265	6.54887
m	392.000	392.000
dfr	7.00000	8.00000
df	384.000	384.000
fStat	146.255	640.418
aic	-1105.39	-821.212
bic	-1073.62	-785.471

Table 14: PyTorch: - Auto MPG In-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.8232	0.8855
rSqBar	0.8199	0.8835
sst	23818.9935	23818.9935
sse	4211.9875	2726.1212
sde	3.3119	2.6644
mse0	10.7449	6.9544
rmse	3.2779	2.6371
mae	2.4941	1.8997
smape	11.4175	8.0857
m	392.0000	392.0000
dfr	8.0000	8.0000
df	384.0000	384.0000
fStat	223.4423	371.3914
aic	946.7758	776.2343
bic	978.5459	808.0044

Table 15: Scalation: - Auto MPG Out-of-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.836932	0.868516
rSqBar	0.833959	0.865777
sst	4731.23	4731.23
sse	771.514	622.081
sde	3.16498	2.79802
mse0	9.89121	7.97540
rmse	3.14503	2.82407
mae	2.36625	2.03520
smape	10.7664	8.62759
m	78.0000	78.0000
dfr	7.00000	8.00000
df	384.000	384.000
fStat	281.549	317.064
aic	-184.051	-173.675
bic	-165.198	-152.464

Table 16: PyTorch: - Auto MPG Out-of-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.8391	0.9095
rSqBar	0.8232	0.9005
sst	4910.2699	4910.2699
sse	790.2830	444.5322
sde	3.3363	2.5022
mse0	10.0036	5.6270
rmse	3.1628	2.3721
mae	2.4081	1.6043
smape	11.1501	7.0419
m	79.0000	79.0000
dfr	8.0000	8.0000
df	71.0000	71.0000
fStat	46.2681	89.1576
aic	197.9325	152.4784
bic	216.8881	171.4340

5.2 California House Prices

Table 17: Scalation: - California House Prices In-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.633776	0.779769
rSqBar	0.633579	0.779639
sst	2.72264e+14	27226.4
sse	9.97097e+13	5996.11
sde	69653.2	0.541323
mse0	4.87984e+09	0.293452
rmse	69855.8	0.541712
mae	51821.3	0.371532
smape	26.8854	19.3019
m	20433.0	20433.0
dfr	11.0000	12.0000
df	20421.0	20421.0
fStat	3212.73	6025.36
aic	-256883	-16441.3
bic	-256788	-16338.3

Table 18: PyTorch: - California House Prices In-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.6468	0.8057
rSqBar	0.6466	0.8056
sst	272264434648930.1875	272264434648930.1250
sse	96172625731084.4375	52901444882088.6094
sde	68625.7706	50897.3609
mse0	4706730569.7198	2589019961.9287
rmse	68605.6162	50882.4131
mae	49654.9430	34838.6035
smape	26.7647	18.1115
m	20433.0000	20433.0000
dfr	12.0000	12.0000
df	20421.0000	20421.0000
fStat	3115.8995	7056.5363
aic	455113.0754	442899.9831
bic	455208.1743	442995.0819

Table 19: Scalation: - California House Prices Out-of-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.630655	0.756155
rSqBar	0.630456	0.756012
sst	5.31945e+13	5319.45
sse	1.96471e+13	1297.12
sde	69346.2	0.562612
mse0	4.80840e+09	0.317454
rmse	69342.6	0.563431
mae	50737.2	0.389204
smape	26.5678	20.5293
m	4086.00	4086.00
dfr	11.0000	12.0000
df	20421.0	20421.0
fStat	3169.89	5277.08
aic	-51319.7	-3427.61
bic	-51243.9	-3345.51

Table 20: PyTorch: - California House Prices Out-of-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.6516	0.7961
rSqBar	0.6506	0.7956
sst	54723646809444.8438	54723646809444.8281
sse	19066274074331.2266	11156430483920.8633
sde	68402.0487	52323.7456
mse0	4665102538.3732	2729735865.8970
rmse	68301.5559	52246.8742
mae	50131.4253	35574.6590
smape	26.6343	18.1492
m	4087.0000	4087.0000
dfr	12.0000	12.0000
df	4075.0000	4075.0000
fStat	635.0821	1326.1142
aic	91014.4163	88824.1727
bic	91090.2031	88899.9595

5.3 Insurance Charges

Table 21: Scalation: - Insurance Charges In-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.722979	0.867231
rSqBar	0.721521	0.866432
sst	1.96074e+11	5446.51
sse	5.43167e+10	723.127
sde	6346.14	0.735248
mse0	4.05954e+07	0.540454
rmse	6371.45	0.735156
mae	4519.55	0.416711
smape	39.5992	24.6632
m	1338.00	1338.00
dfr	7.00000	8.00000
df	1330.00	1330.00
fStat	495.869	1085.92
aic	-13602.9	-1468.87
bic	-13561.3	-1422.08

Table 22: PyTorch: - Insurance Charges In-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.7516	0.8766
rSqBar	0.7503	0.8760
sst	196074221568.3671	196074221568.3671
sse	48702200251.0962	24191406710.2515
sde	6051.2970	4264.8596
mse0	36399252.8035	18080274.0734
rmse	6033.1793	4252.0906
mae	4158.4738	2466.8147
smape	39.1264	23.6520
m	1338.0000	1338.0000
dfr	8.0000	8.0000
df	1330.0000	1330.0000
fStat	503.0696	1181.2260
aic	23310.6587	22374.4243
bic	23352.2501	22416.0158

Table 23: Scalation: - Insurance Charges Out-of-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.682065	0.817677
rSqBar	0.680392	0.816580
sst	4.06432e+10	1128.98
sse	1.29219e+10	205.838
sde	6966.63	0.877561
mse0	4.83965e+07	0.770930
rmse	6956.76	0.878026
mae	4729.99	0.534961
smape	41.9288	31.2906
m	267.000	267.000
dfr	7.00000	8.00000
df	1330.00	1330.00
fStat	407.607	745.594
aic	-2725.13	-326.127
bic	-2696.43	-293.842

Table 24: PyTorch: - Insurance Charges Out-of-Sample QoF Comparison

Metric	2 Layer NN	3 Layer NN
rSq	0.8007	0.8891
rSqBar	0.7953	0.8861
sst	42646829902.3297	42646829902.3297
sse	8498989265.5920	4730156280.8836
sde	5717.3788	4265.3146
mse0	31712646.5134	17649836.8690
rmse	5631.3983	4201.1709
mae	3932.9280	2735.5215
smape	36.6337	27.4890
m	268.0000	268.0000
dfr	8.0000	8.0000
df	260.0000	260.0000
fStat	130.5808	260.5182
aic	4644.9566	4487.9115
bic	4673.6845	4516.6394